

Introduction of rough paths

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- 1 controlled differential equations
 - How to solve a controlled ODE?
 - How to extend the rough path theory of controlled ODEs to PDE?
 - application and difference from other textbook
- 2 Analogies with other branches of mathematics
- 3 Regularity structure(Ref. ch14, ch15)
- 4 Frequently used notations

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How to solve the following controlled ODE?

$$\dot{Y}_t = f_0(Y_t) + f(Y_t)\dot{X}_t, \quad (1)$$

with the initial state Y_0 . $X \in \mathbf{R}^d$ is the input, $Y \in \mathbf{R}^e$ is the output, f_0 and f are nonlinear functions.

If $\dot{X} = \xi$ is the white noise in time ($X \in B$ is a Brownian motion), we need a non-smooth theory!

Solving methods

- Discrete method
- Probabilistic techniques method
- Rough path analysis method-introduced [Lyo98']

Approximate (1) as the scaling limit ($h \rightarrow 0$) of the following discrete evolution equation

$$Y_{i+1} = Y_i + hf_0(Y_i) + \sqrt{h}f(Y_i)\xi_{i+1} \quad (2)$$

(ξ_i) i.i.d. standard Gaussian random variables.

The martingale theory + Itô's SDEs $\Rightarrow$

the solution map $(\xi_1, \xi_2, \dots, \xi_T; Y_0) \rightarrow Y_i$ is surely a continuous map.

However, the stability is lost in the passage to continuous time !

The solution map (Itô's map) $S : B \rightarrow Y$ to (1) is a measurable map which in general lacks continuity, whatever norm one uses to equip the space of B !

Proposition

[Lyo91, LCL07, Ex.5.21] *There exists no separable Banach spaces $\mathcal{B} \subset \mathcal{C}([0, 1])$ with the following properties:*

- 1. Sample paths of Brownian motions lie in $\mathcal{B}$ a.s..*
- 2. The map $(f, g) \rightarrow \int_0^\cdot f(t)\dot{g}(t)dt$ defined on smooth functions extends to a continuous map from $\mathcal{B} \times \mathcal{B}$ into the space of continuous functions on $[0, 1]$.*

Example: $Y \in \mathbf{R}^2$,

$$\begin{aligned}\dot{Y}^1 &= \dot{B}^i, \\ \dot{Y}^2 &= Y^1 \dot{B}^j,\end{aligned}$$

The solution map:

$$B \rightarrow \int_0^\cdot B^i(t) \dot{B}^j(t) dt, i \neq j, \quad (3)$$

lacks continuity.

Theorem

[INY78'] Let $\xi_\epsilon = \delta_\epsilon * \xi$ denote the regularization of white noise in time with a compactly supported smooth mollifier δ_ϵ . Denote by Y^ϵ the solutions to (1) driven by $\dot{X} = \xi_\epsilon$. Then Y^ϵ converges in probability (uniformly on compact sets). The limiting process does not depend on the choice of mollifier δ_ϵ , and in fact is the Stratonovich solution to

$$dY_t = f_0(Y_t)dt + f(Y_t) \circ dB_t, \quad (4)$$

here all coefficients are assumed to be sufficiently nice.

Remark

- 1: *It turns out by [INY78'] that well-posedness is restored via the iterated integrals (3) using the probability statements. (actually the integral is missing to turn S into a continuous map).*
- 2: *Another popular choice being $\xi_\epsilon = \dot{B}^{(\epsilon)}$ where $B^{(\epsilon)}$ is a piecewise linear approximation to Brownian motion.*
- 3: *We also have explicit approximations(Ref. §3.4) to white noise for which the above convergence fails.*

$$\dot{Y}_t = f_0(Y_t) + f(Y_t)\dot{X}_t$$

Key point

Itô's solution map can be factorized into a measurable "universal" map Ψ and a "nice" solution map $\hat{S}$:

$$B(\omega) \xrightarrow{\Psi} (B, \mathbb{B})(\omega) \xrightarrow{\hat{S}} Y(\omega). \quad (5)$$

- The map Ψ is universal

Depends neither on the initial condition, nor on the vector field driving the stochastic differential equation. It only consists of enhancing Brownian motion with iterated integrals of the form

$$\mathbb{B}^{i,j}(s, t) = \int_s^t (B^i(r) - B^i(s)) dB^j(r). \quad (6)$$

- How to construct the Ψ ?

Careful choice of stochastic integration in (6) and probabilistic techniques (Ref. ch5 in the Itô or the Stratonovich sense)

The map Ψ is only measurable and usually requires the use of some sort of stochastic integration theory (Ref. ch10 for a general construction in a Gaussian, non-semimartingale context).

- Itô-Lyons map $\hat{S}$: only make use of analytical construction (Ref. §8.1)
It allows input signals to be *arbitrary rough paths*(Ref.ch2)/*Brownian rough paths*(Ref.ch3)- objects of the form $(X, \mathbb{X})$ (or called enhanced paths), defined via certain algebraic properties (which mimic the interplay between a path and its iterated integrals), and certain analytical, hölder-type regularity condition.
"enhanced" means that $\mathbb{X}$ is enhanced with a two-parameter function $\mathbb{X}$ which should be interpreted as giving the values for $\int_s^t X_{s,r}^i dX_r^j := \mathbb{X}^{ij}(s, t)$.

- Property of Itô-Lyons map $\hat{S}$ (Ref. §8.6)
it is continuous both its initial condition and the driving noise $(X, \mathbb{X})$,
provided that the dependency on the latter is measured in a suitable
"rough path" metric (Ref. Th8.5)
Rough path analysis allows for a pathwise solution theory for SDEs
i.e. for a fixed realization of the Brownian rough path.

- How to build the Itô-Lyons map $\hat{S}$?

The construction is independent of the choice of stochastic integral (Itô/ Stratonovich) and even of the knowledge that the driving path is Brownian.

- 1st step (Ref. ch4): make sense of the expression

$$\int_0^t f(Y_s) dX_s, \quad (7)$$

where Y is unknown solution. It is in general impossible to find a Banach space (in the simplest case) of functions containing Brownian sample paths and in which (7) makes sense. It makes sense as a limit of Riemann sums for X and Y that are arbitrary α -Hölder continuous functions if and only if $\alpha > \frac{1}{2}$.

- How to build the Itô-Lyons map $\hat{S}$?
 - 2nd step : "enhance" a path X with some additional data $\mathbb{X}$ (the integral of X) to restore continuity of the Itô's map
enhance the X (e.g. Brownian motion, stochastic process with properties similar to those of Brownian motion) to be a "rough path"

$$\int_s^t X_{s,r}^i dX_r^j := \mathbb{X}^{i,j}(s, t), \quad X_{s,r} := X_r - X_s, \quad (8)$$

(with $\mathbb{X}$ satisfying a number of natural algebraic identities and analytical bounds, Ref. ch2).

Remark

If X is smooth, then we "know" (not automatically) that the only reasonable choice for the integral-Riemann integral (Ex2.17 an explicit example)

Here, "non-canonical" rough paths can also be constructed as limits of ordinary smooth paths (with the second-order term $\mathbb{X}$ defined by (8) where the integral is the usual Riemann integral), provided that one takes limits in a suitably weak topology.

- How to build the Itô's-Lyons map $\hat{S}$?
 - 3rd step: extending the definition of (7) to a large enough class of integrands $\int_0^t f(Y_s) d(X, \mathbb{X})$ to include solutions to (1) by encoding the fact that Y "looks like X at small scales"
 - (i) $Y_{s,t} = f(Y_s)X_{s,t} + R_{s,t}$ (ch 4)
Moreover, the barrier $\alpha = \frac{1}{2}$ can be lowered to $\alpha = \frac{1}{3}$ at least in the "geometric" case when X is a limit of smooth paths (§2.4, §2.5, §7.6).
 - (ii) $Y = G(X)$ for some sufficiently regular function G [Lyo98']
 - (iii) $Y_{s,t} = Y'_s X_{s,t} + R_{s,t}$, with a remainder satisfying $R_{s,t} = O(|t - s|^{2\alpha})$ [Gub04']

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How to extend the rough path theory of controlled ODEs to PDE?

- finite dimensional noise—Theorem 1.3 (Ref. §12)

Theorem

Classes of SPDEs of the form $du = F[u]dt + H[u] \circ dB$ driven by finite-dimensional noise, F and H are second and first order differential operators respectively (e.g. the Zakai equation from filtering and stochastic Hamilton-Jacobi-Bellman (HJB) equations) can be solved pathwise, i.e. for a fixed realization of the Brownian rough path. As in the SDE case, the SPDE solution map factorises as $S^s = \hat{S} \circ \Psi^S$ where $\hat{S}$, the solution map to a rough partial differential equation (RPDE) is continuous in the rough path topology.

How to extend the rough path theory of controlled ODEs to PDE?

Remark

1: As a consequence, if we regularize the white noise in time with a compactly supported smooth mollifier δ_ϵ : $\xi_\epsilon = \delta_\epsilon * \xi$, and we denote u^ϵ as the random PDE solutions driven by $\xi_\epsilon dt$ (instead of $\circ dB$), then u^ϵ converges in probability. This limiting process does not depend on the choice of mollifier δ_ϵ , is viewed as Stratonovich SPDE solution.

2: The same conclusion holds whenever $\Psi^S(B^\epsilon) \rightarrow \Psi^S(B)$ in probability and rough path topology.

How to extend the rough path theory of controlled ODEs to PDE?

- 1 space dimension:
 - linear case: Feynman-Kac approach(make use of concentration of measure result);
 - nonlinear case: flow transformation method, focus on three types of noise
- higher space dimension: needs the general theory of regularity structures.

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- application

Other results using the rough paths theory: quantum field theory, nonlinear PDEs, Malliavin calculus (§11.3), non- Markovian Hörmander theory, ergodic theory, the analysis of chaotic behavior in fast-slow systems.

- difference from other textbook

In this book, the authors emphasize Gubinelli's view on rough integration, which allows to linearise many considerations and to simplify the exposition.

Main results: the definition and continuity the Itô's-Lyons map, the lifting of a class of Gaussian processes to the space of rough paths, ect.

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Analogies with other branches of mathematics

- *theory of rough paths VS Young measure*

The theory of rough paths: building a larger object containing additional information in order to restore the continuity of some nonlinear transformation

Young measure: replace the value of a function by a probability measure, to describe the limits of highly oscillatory functions

- *theory of rough paths VS varifolds* (geometric measure theory)
The theory of rough paths: "non-canonical" rough paths can be constructed as limit of ordinary smooth paths(with the second-order term $\mathbb{X}$ defined by (8) where the integral is the usual Riemann integral), provided that one takes limits in a suitably weak topology.
Varifolds: extensions of submanifolds in the context of certain variational problem

Analogies with other branches of mathematics

Remark

*a k dimensional varifold in $\mathbf{R}^n$ is a Randon measure $\mathbf{v}$ on $\mathbf{R}^n \times \mathcal{G}(k, n)$
 $\mathcal{G}(k, n)$ denotes the space of all k -dimensional subspaces of $\mathbf{R}^n$ (the space of all possible tangent spaces at any given point for a k -dimensional submanifold of $\mathbf{R}^n$)*

The projection of $\mathbf{v}$ onto $\mathbf{R}^n$ is a generalization of the natural "surface measure" of a submanifold, while the probability measure on $\mathcal{G}(k, n)$ induced at almost every point by disintegration is interpreted as selecting a possibly random tangent space at each point.

Analogies with other branches of mathematics

- 1st example in varifolds

the sequence $\{M_\epsilon\}$ of one dimensional submanifolds of $\mathbf{R}^2$ converges to a circle, as $\epsilon \rightarrow 0$. The right half has twice as much "weight" as the left half. We would lost some information about the convergence of the surface measures if we just describe the limit M simply as a manifold.

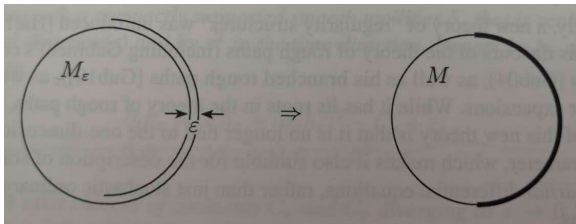


Figure: 1st example

Analogies with other branches of mathematics

- 2nd example in varifolds

The limit of the sequence of one dimensional submanifolds of $\mathbf{R}^2$ is a piece of straight line, which is a perfectly nice smooth submanifold, but the limiting tangent space is deterministic and makes a 45° angle with the canonical tangent space associated to the limit.

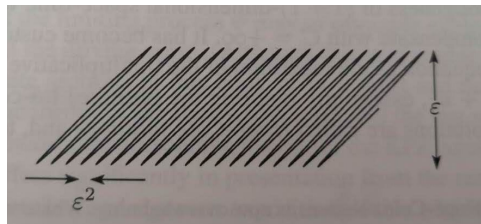


Figure: 2nd example

A sequence of smooth manifolds such that limit is again a smooth manifold, but with a limiting "tangent space" which has nothing to do with the actual tangent space of the limit!

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Regularity structure (Ref. ch14, ch15)

- aim: A unified theory suitable both for stochastic ODEs and stochastic PDEs
- main achievement : when we describe the macroscopic behavior of models from statistical mechanics near criticality, this theory allows to give a pathwise meaning to ill-posed stochastic PDE.

ch14 will give a short and self-contained introduction to the theory of regularity structures

ch15 shows how it can be used to provide a robust solution theory for the KPZ equation

Regularity structure(Ref. ch14, ch15)

- examples

KPZ equation (model for one-dimensional interface motion):

$$\partial_t h = \partial_x^2 h + (\partial_x h)^2 - C + \xi, \quad (9)$$

Dynamical Φ_3^4 model(model in the stochastic quantization of Euclidean quantum field theory, or for phase coexistence near the critical point):

$$\partial_t \Phi = \Delta \Phi + C\Phi - \Phi^3 + \xi. \quad (10)$$

The theory of regularity structures can help us to get the well-posedness results:

Theorem

Consider KPZ and Φ_3^4 on a bounded square spatial domain with periodic boundary conditions. Let $\xi_\epsilon = \delta_\epsilon * \xi$ denote the regularisation of space-time white noise with a compactly supported smooth mollifier δ_ϵ that is scaled by ϵ in the spatial direction and by ϵ^2 in the time direction. Denote by h_ϵ and Φ_ϵ the solutions to

$$\begin{aligned}\partial_t h_\epsilon &= \partial_x^2 h_\epsilon + (\partial_x h_\epsilon)^2 - C_\epsilon + \xi_\epsilon, \\ \partial_t \Phi_\epsilon &= \Delta \Phi_\epsilon + C_\epsilon \Phi_\epsilon - \Phi_\epsilon^3 + \xi_\epsilon.\end{aligned}$$

Then, there exist choices of constants C_ϵ and $\tilde{C}_\epsilon$ diverging as $\epsilon \rightarrow 0$, as well as processed h and Φ such that $h_\epsilon \rightarrow h$ and $\Phi_\epsilon \rightarrow \Phi$ in probability. Furthermore, while the constants C_ϵ and $\tilde{C}_\epsilon$ do depend crucially on the choice of mollifiers δ_ϵ , the limits h and Φ do not depend on them.

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Frequently used notations

- Tensor products in finite Banach spaces

- definition

$$V \cong \mathbf{R}^m, V \cong \mathbf{R}^n, V \otimes W \cong \mathbf{R}^{m \times n}$$

its basis: $\{e_i \otimes f_j : e_i, f_j \text{ is the base of } V(W)\}$.

- the property:

1 . orthonormal base $\{e_i \otimes f_j\}$, if (e_i) and (f_j) are orthonormal bases,

$$|v \otimes w| \leq |v||w|, \forall v \in V, w \in W;$$

2 . permutation invariance property

$$|u \otimes v| = |v \otimes u|, \forall u, v \in V;$$

3. linear bilinear map property

$$\mathcal{L}(V \times \bar{V}, W) \cong \mathcal{L}(V \otimes \bar{V}, W), \mathcal{L} \text{ is the linearize bilinear map.}$$

Frequently used notations

- nonlinear map between Banach spaces

$$\|F\|_{\mathcal{C}_b^n} \equiv \|F\|_\infty + \|DF\|_\infty + \cdots + \|D^n F\|_\infty < \infty, \text{ in Fréchet sense,}$$

where $F \in \mathcal{C}_b^n$, $DF(v) \in \mathcal{L}(V, W)$,
 $D^2F \in \mathcal{L}(V, \mathcal{L}(V, W)) \cong \mathcal{L}(V \times V, W)$

- the α hölder semi-norm:

$$\|X\|_\alpha := \sup_{s,t \in [0,T]} \frac{|X_{s,t}|}{|t-s|^\alpha} < \infty,$$

$X_{s,t} := X_t - X_s$ is the path increment.

We will work with $\|\cdot\|_\alpha$ by assuming that the path at the origin.

Frequently used notations

- denote $\Xi(s, t) := \Xi_{s,t} \in \mathcal{C}_2^\alpha$ if there exists a constant C such that $|\Xi_{s,t}| \leq C|t - s|^\alpha$ for all $(s, t) \in [0, T]^2$ (the smallest constant is given by $\|\Xi\|_\alpha$). example: if $X \in \mathcal{C}^\alpha[0, T]$, then $X_{s,t} \in \mathcal{C}_2^\alpha[0, T]^2 \dots$
- other symbols: $\mathcal{C}^\alpha$, $\mathcal{D}_X^\alpha$ refer to spaces of rough paths and controlled rough paths.

The End